AI for Making Cultural Heritage Accessible at the Royal Dramatic Theatre

Using a Retrieval-Augmented Generation Approach

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Abstract

Introduction This thesis studies the viability of making cultural heritage easily accessible through AI by using the RAG (Retrieval-Augmented Generation) framework on a locally deployed LLM. The study explores the Royal Dramatic Theatre's vast archive of digital and physical material.

Research Question The primary research question is: "How can a locally run AI model make the Royal Dramatic Theatre's archive more easily accessible?"

Method 12 queries about the play *Hamlet* were prompted into the developed RAG system prototype to see if it could answer them using the information within the archive. Two groups of different sizes of segmented data known as *Chunks* were tested on the 12 queries to see how it affected the system in finding and interpreting the needed context to each query.

Results The results showed that a smaller chunk size for structured data gives better responses to fact based queries. Open queries were not as sensitive to different chunk sizes and performed similarly between the two groups of chunk sizes. The system was overall able to answer most queries using the archive, which suggests the potential of making cultural heritage accessible with such a system.

Discussion The findings showed limitations within the prototype, such as misinterpreting the context from the archive to produce wrong answers or not finding relevant context at all. Further research could study the use of metadata and improvement of system instructions to reduce AI hallucinations. Query transformation and spellchecking could also be explored to help the system interpret queries better and find the correct information within the archive.

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
OCR	Optical Character Recognition
TDM	Text and Data Mining
CDSM	Directive on Copyright in the Digital Single Market
GDPR	General Data Protection Regulation

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1

Introduction

Cultural heritage is constantly being archived, and in the age of technology, digital archives are becoming increasingly common. Older material tends to be originally archived in physical form as a consequence of technological limitations during those ages. There are different organizations such as the Swedish National Heritage Board that have the mission to preserve such artifacts and part of their goal is to digitalize cultural heritage (Riksantikvarieämbetet, 2026).

The use of metadata for digital material is important for things like discoverability, integrity and authenticity (Selva, 2025). One common mistake that weakens archives is manual only workflows, which can be things such as manually tagging archived material with metadata, as humans can easily make mistakes (Selva, 2025). Physical material can come in many forms such as images and texts which doesn't come with any metadata and need to be digitalized and tagged in order to make it more accessible. With large sets of physical material, this becomes a challenge that needs to be addressed.

As the use of AI technology is rising, it has been proposed as a method of cultural preservation. In a report by UNESCO (United Nations Educational, Scientific and Cultural Organization), the opportunities where AI can assist in digital preservation, virtual reconstruction and data analysis are discussed. An issue with AI is that many of those systems are trained on large datasets which are collected through web scraping without the

owner's knowledge or consent (UNESCO, 2025). UNESCO has taken initiative in response to this, however, there are still legal loopholes that make safeguarding cultural rights difficult. Some datasets rely solely on metadata or hyperlinks instead of storing the entire work, where there is no legal protection to partially stored work being used without knowledge or consent (UNESCO, 2025).

This study explores the Royal Dramatic Theatre's large archive of cultural heritage that is partly digital and partly physical. It is currently inaccessible to the public and should be made easily accessible to those interested by utilizing AI in a way where cultural rights also remain safeguarded.

1.1 Problem

The archive at the Royal Dramatic Theatre remains difficult to access. Although parts of the archive are digitally available, a significant portion still exists only in physical formats, making it difficult to access and explore. Archival data collections are often characterized by fragmented metadata and require prior knowledge to navigate effectively (Kelly et al., 2025).

With the use of generative AI models, new possibilities for improving access to cultural heritage through natural language interaction arise. However, the use of cloud-based AI models raises legal concerns when discussing copyrighted material. The European Union Article 4 of The CDSM (Copyright in the Digital Single Market) currently has no protection over such data being used by generative models. Because of this, authors and organizations must let their work be used or forcefully decline by using TPMs (technological protection measures) (Lucchi, 2025). EU law does provide protection from circumventing these TPMs (Lucchi, 2025), however, this means that you are forced to use them instead of having your copyrighted work be protected by default. Having a locally deployed solution where the copyrighted data being used is stored by the owners becomes a viable option as a result. This also means that the implementation can't be too resource consuming as there is no dependency on powerful cloud-based AI models.

AI models are prone to generating incorrect or misleading information, also known as hallucinations. This is problematic when dealing with historical content, where accuracy and trustworthiness are important. Also,

pre-trained language models rely on knowledge learned during training, which makes it difficult to update or add new information (Lewis et al., 2021). Ensuring accurate, up-to-date, and source-grounded responses is therefore a key challenge.

1.2 Research Question

How can a locally run AI model make the Royal Dramatic Theatre's archive more easily accessible?

2

Extended Background

2.1 Retrieval-Augmented Generation for Cultural Heritage Access

Large language models (LLMs) can generate human-like text and answer complex user queries. However, LLMs rely on knowledge learned during training and do not have access to updated or external information (Lewis et al., 2021). As a result, these systems may generate incorrect information, a phenomenon known as hallucination, which occurs when an AI system fabricates responses that deviate from factual data but remain linguistically convincing (Nambiar & Sreedevi, 2023). In addition, LLMs typically do not provide sources of their claims, making it difficult to verify the accuracy.

A possible solution to this is to use Retrieval-Augmented Generation (RAG). RAG is an approach that uses both parametric and non-parametric memory for generating text. Parametric memory is the knowledge learned during training, while non-parametric memory is data stored externally, such as a database or text document (Lewis et al., 2021). This approach makes it possible to update the knowledge of the AI system without having to retrain it, reduces the risk of hallucinations, and increases transparency (Lewis et al., 2021). Within the cultural heritage domain, RAG enables different archival materials to be accessed using natural language queries (Kelly et al., 2025).

2.1.1 The RAG Workflow

The RAG workflow is typically divided into three stages: retrieval, augmentation, and generation. The process begins with a user query, which is used to retrieve relevant text segments from the non-parametric memory based on their semantic similarity to the query (Lewis et al., 2021). In the next step, the prompt is augmented by combining the retrieved text as context with the original query. Finally, the AI model generates a response based both on the provided context and on its parametric knowledge (Lewis et al., 2021). This ensures that the final output is based on verifiable external sources.

2.1.2 Semantic Search and Vector Embeddings

Unlike traditional search methods that rely on keyword matching, RAG typically uses semantic search (Reddy K et al., 2025). In semantic search, text data is transformed into vector embeddings, which are numerical representations of text segments. These vector embeddings enable similarity-based vector search, where the relationship between a query and external documents is calculated, often using measures such as cosine similarity (Kelly et al., 2025). As a result, the system can identify relevant information based on semantic meaning rather than exact word matches. For example, a query such as "Who played Hamlet?" can retrieve documents containing the term "actor", even though the exact words are different because the underlying meaning is similar. This is particularly useful when working with historical archives, as the language can change over time. This means that the system can understand and use old words in the archive and connect them to user queries with modern terminology.

2.1.3 Challenges and Limitations of RAG

Despite its advantages, RAG systems face challenges. Like standard LLMs, they remain susceptible to bias if the external source material is biased or factually incorrect (Lewis et al., 2021). Furthermore, the process of dividing long documents into smaller segments, also called data chunking, impacts the system precision. For example, too large text chunks can dilute semantic specificity, making it harder for the model to retrieve the most relevant information (Kelly et al., 2025). Therefore, the effectiveness of a RAG system is highly dependent on how external data is processed, structured, and segmented prior to retrieval.

2.1.4 RAG in Cultural Heritage and Archival Systems

Recent research has explored the use of RAG in the context of cultural heritage and archival systems. According to Kelly et al. (2025), the FolkRAG system was developed to address the challenges in accessing archival material from the American Folklife Center. In this context, fragmented metadata and disconnected data sources, such as catalog records, finding aids and digital objects, make navigation difficult.

FolkRAG integrates LLMs with external retrieval mechanisms to enable natural language queries over archival data, providing responses grounded in source material to ensure reliability and transparency (Kelly et al., 2025). A central technical finding from the FolkRAG study is the importance of document chunking strategies. The study found that 250-character chunks provided better results compared to larger chunk sizes. This is because archival material often consists of information-dense segments, where smaller chunks help isolate this information instead of diluting it, resulting in more relevant content for retrieval.

The FolkRAG study demonstrates how RAG can provide a more intuitive interface for exploring archival material. This highlights the relevance of RAG-based approaches for the system proposed in this study and supports their applicability in similar archival contexts.

2.2 Locally Hosted LLM for Cultural Heritage Access

A locally hosted LLM within this study's context is the concept of running an AI model on your own hardware rather than using a third party's. The advantages and challenges of using locally hosted LLM as opposed to cloud-based LLMs such as open AI's ChatGPT and Google's Gemini can be discussed from different perspectives.

2.2.1 Data Privacy for Copyrighted Material

AI models require large amounts of training data in order to provide more accurate outputs. This can become an issue when training or retrieving context from copyrighted material. To understand the state of the law in Europe regarding training on copyrighted material, TDM must be discussed. TDM (Text and data mining) which is defined in the CDSM (Directive on

Copyright in the Digital Single Market) as:

"Any automated analytical technique aimed at analysing text and data in digital form in order to generate information which includes but is not limited to patterns, trends and correlations." (Lucchi, 2025).

In the context of copyrighted material, generative AI is often discussed as it doesn't just extract existing data and instead produces new content that can contend with original work (Lucchi, 2025). It has been argued that this is a reason why generative AI does not technically or legally count as TDM (Lucchi, 2025). This is important as the CDSM have exceptions to what TDM can be used for in order to train generative AI, which includes "the reproduction and extraction of protected works for TDM purposes" (Tristan & Marcel, 2025).

With all these legal loopholes and vulnerabilities in protecting copyrighted work, the use of locally hosted AI models becomes more interesting for making cultural heritage more accessible. Instead of relying on third party AI companies to train on copyrighted material, the information could be stored somewhere where only your locally hosted LLM can access it.

2.2.2 Resource Limitations of Hosting Local LLMs

The choice of locally hosting an LLM can also be a result of resource limitations. For example, the state of the art model GPT-4o required 38 billion petaFLOPs for training, with 1 petaFLOP being 10^{15} floating point operations (Stanford, 2025). While these are more advanced, smaller models have arisen to make computing on less hardware more feasible. An example of this is Mistral which comes in different variations. One version called mistral 7b scored above 60% on MMLU (Stanford, 2025). Massive multi task language understanding (MMLU) is a benchmark for assessing LLM performance through multiple choice questions within 57 subjects (Stanford, 2025).

Number of parameters also affect resource consumption with higher parameter values increase the amount of data needed to train a model with the advantage of being able to perform more tasks. Parameters are learned during training and determine how models make predictions and interpret input data (Stanford, 2025).

There are even smaller versions of Mistral that run on less resources with different degrees of quantization. Quantization is an optimization technique used by LLMs where high precision data like 32-bit floating point (FP32) or FP16 is converted to a lower precision data (Clark, 2026). This reduces the accuracy of output with the advantage of increasing efficiency through less computation time, power and energy consumption.

When making cultural heritage easily accessible through LLMs the option of data privacy can come at a cost of accuracy when hosting locally. This is true assuming that whoever wants to achieve this doesn't have a similarly large infrastructure as third parties have.

2.2.3 The Study of Locally Hosted LLMs to Increase Search Ability of Archives

The use of locally hosted large language models to make archives easier to access has been brought up before. In (Reusens et al., 2025) Meta's open source language model Llama was used to make the National Museum of the Royal Navy's archive more accessible. This was done through Llama's ability to generate keywords in order to increase the search ability of the archive (Reusens et al., 2025). Reusens et al. (2025) explains that the use of LLMs like Llama made generating meta data for material easier which also made it easier to make the material more searchable. The dataset included metadata for each item such as description, number, associated subject, person and date in order for the model to generate keywords. An important point that Reusens et al. (2025) brought up is how the open source nature of Llama made it more cost effective and also increased data privacy as it could be operated independently of big tech companies.

3

Method

3.1 Research Strategy

This study follows a design science research methodology, an approach suited for solving practical problems by creating and evaluating an artifact. In this case, the artifact is a RAG-based chat system connected to the Royal Dramatic Theatre's archive, allowing users to interact with archival material through natural language queries.

The study employs an iterative process consisting of three main phases:

1. **Problem identification**: This phase involves finding and defining the challenges associated with accessing historical archival material and data protection regulations. This was discussed in chapter 1.1.
2. **Design and development**: This phase focuses on the technical implementation of the artifact, including processing of the archival data, integration of the AI model, and development of the RAG pipeline.
3. **Evaluation**: In this phase, the performance of the artifact is evaluated based on its ability to provide accurate responses. This also happens iteratively throughout the process of implementing the artifact.

3.2 System Design

The system is a RAG-based chat application that users interact with to make the Royal Dramatic Theatre's vast archive more accessible through queries. This study focuses on queries regarding the play Hamlet.

3.2.1 Tools

This is a list of tools that were used when designing the system.

Python: The system was written with the Python programming language.

Mistral Small 3.2: The open source model Mistral Small 3.2 was used as the LLM.

Ollama: Mistral Small 3.2 was run locally using Ollama.

all-MiniLM-L6-v2: Was used as the embedding model which turns data into numerical vectors, which are later used for semantic search

PaddleOCR: An Optical Character Recognition engine that transcribes text inside images into a markdown file.

Streamlit: A python framework for building interactive web applications which has built in functions for creating an LLM chatbot demo.

3.3 Data Acquisition

The archival material from the Royal Dramatic Theatre was categorized into digital and physical sources.

3.3.1 Digital Data

The digital data used consisted of metadata regarding productions, roles, and cast lists retrieved from their website, *Rollboken* (the rolebook). Instead of scraping rendered HTML, the data were collected through publicly accessible web API endpoints.

The *Rollboken* website contains a list of Hamlet productions from different years, each entry with a unique *play ID*. These ID:s were used to retrieve detailed information about each production through separate API endpoints. For each production, three types of data were collected:

- General production metadata (e.g. title, premiere date, stage, number of performances)
- Production staff information (e.g. director, scenographer, author)
- Cast information (actor, role, performance date)

The data were retrieved via HTTP GET requests and returned in JSON format. All collected data were stored locally in a JSON file with a structured object for each production containing metadata, production roles and cast.

3.3.2 Physical Data

The physical data consisted of newspaper articles from year 1960 about the Hamlet play at the Royal Dramatic Theatre. There was information about the people working on the set, the people playing the roles, reviews from various sources within the articles and other descriptive information. A subset of these newspaper articles was photographed to be transcribed by PaddleOCR which is an OCR (Optical Character Recognition) to create a markdown file for each image with the article text. Since one page could contain many different articles, most individual articles were edited by zooming in on them to make sure only the specific article was transcribed into text. Zooming in also made the characters larger and more visible, which helped making the transcription process more accurate. Bigger articles were divided into multiple images to fit all the content. The choice of PaddleOCR was for its ability to understand the structure of each article and make sure columns were properly recognized.

3.4 Implementation

Queries were sent through a chat application to a local instance of Mistral Small 3.2 running with Ollama. The choice of Mistral Small 3.2 was due to the low computing power it requires to run compared to other Mistral models, while not being too simple. The latest version of Mistral Small is 4.0 which is more advanced but also requires a lot more computing power to run. Using version 3.2 can be a small sacrifice to the accuracy of the

answers but with the benefit of being easier to run locally by using less resources to achieve data privacy. Mistral was also used to explore the viability of European LLM alternatives. To make answers more accurate and avoid AI hallucinations, the RAG framework was used, where extra context to each query was retrieved from the Royal Dramatic Theatre's archive. The model was also provided with a set of system instructions that specified that the model should answer based on context and not fabricate information. The chat interface was built using Streamlit, a Python framework for creating web-based applications.

3.4.1 Context Processing

Context was derived from images and web scraped data using the RAG framework. The raw text data from both sources were split into fixed size chunks. These chunks were embedded to give them numerical values that represent their positions in the produced vector database. Each query being sent were also embedded where a similarity search was made to retrieve the most relevant chunks of data to the query to be used as context. More precisely it would be the chunks with the similarity scores closest to 1.0. The retrieved context was combined with the user's queries to provide a more accurate answer. The top 8 chunks by similarity score were retrieved for each query to be used as context.

3.5 Evaluation Method

3.5.1 Data Collection

A set of 12 Hamlet based queries were sent to the local RAG implemented LLM where the answers were collected. 6 of them were fact based while the other 6 were open queries. The queries covered things that required the information from the Royal Dramatic Theatre's archive. This was done to make sure the system retrieved and interpreted the context correctly with each query.

3.5.2 Data Analysis

A comparison of different chunking sizes and how they affect context retrieval, interpretation and system response were made. The study also compared how well different chunk sizes performed on factual and open queries. This is also interesting as the rolebook data mainly contained structured data while the newspaper articles contained natural language which can

give insight on how the type of query affects the system's handling of context. The responses were classified in the following way:

Factual response: The system was able to find the correct context from the archive and use it to answer the query

Retrieval failure: The system was not able to find the correct context from the archive even though it exists, which prevents the system from answering the query

Contextual misrepresentation response: The system was able to find context that is within the queries theme but not relevant to the specific query. This causes the system to assume that the context is relevant and starts giving hallucinated answers instead of abstaining from answering the query.

3.6 Alternative System Design Methods

3.6.1 Keyword Search

An alternative to the RAG approach would be to store all data in a relational database and retrieve the relevant information via SQL queries or keyword search, instead of semantic search. While this method can be effective in some cases, for example for structured data, it is not as effective for unstructured data, such as newspaper reviews. Traditional SQL queries require exact word matches, which make them less capable of handling the semantic nuance of historical texts.

3.6.2 Fine-Tuning

Another alternative method is fine-tuning, where a pre-trained LLM is further trained on a specific dataset. Both RAG and fine-tuning aim to improve the system knowledge on a specific domain, and they are often combined to produce optimal results. Fine-tuning can be useful when you want the AI model to act and respond in a certain way or do specialized tasks, since the underlying LLM is modified. Fine tuning is more resource intense since it requires LLM training, while RAG is only a retrieval mechanism. This also makes handling updates in the archival material more efficient with RAG. The RAG approach is better at avoiding hallucinations since it grounds its

answers on retrieved documents. Therefore, the RAG approach is better suited for this study.

3.7 Alternative System Evaluation Methods

It might be difficult to study how AI can be used to make a cultural archive easily accessible using anything but design science. There are however other methods to consider when evaluating the prototype. One suggestion is to let people who work with the archive at the Royal Dramatic Theatre test the prototype. A semi structured interview could take place afterwards, where questions about the prototype's accuracy could be asked to explore the viability of such a system. This group of participants is more suitable for such questions as they have good knowledge of the archived material.

Another group of people that could be studied instead are theatre enthusiasts, where the same principle applies. The prototype is tested and afterwards a semi structured interview is conducted. Questions about the user's experience in using such as system and their attitudes towards it could be a bigger focus with such a group of people. This would allow the research question to be answered from a different point of view as opposed to the system's accuracy.

Semi structured interviews are more viable than a quantitative method like surveys regardless of the types of participants for the evaluation. They would allow the measurement of opinions, feelings, emotions and experiences when using the system (Denscombe, 2017). Other thoughts and ideas which aren't covered by the interviewer's questions can be raised by the interviewees to bring other aspects into the discussion (Denscombe, 2017).

3.8 Ethics

An ethical concern when using LLMs for historical archives is the risk of AI hallucinations, when the system generates plausible but factually incorrect information. This is particularly problematic when dealing with cultural heritage at a state cultural institution. To reduce this risk, the system is designed using a RAG architecture. By instructing the model to rely strictly on the provided archival material and explicitly state when information is unavailable, the systems aims to reduce fabrication. However, as there is no

guarantee of the system being free from error, transparency remains important. The system interface therefore clearly informs the user that responses are generated by an AI that is based on archive material.

Data protection regulations such as the General Data Protection Regulation (GDPR) are considered by ensuring that the system runs locally and does not process any sensitive personal data. Unlike cloud-based AI models, where data may be stored by third-party corporations, a local implementation ensures that the archival data does not leave the infrastructure of the Royal Dramatic Theatre.

It is important to acknowledge that historical archives, such as old newspaper articles, may contain cultural biases of its time. The system does not censor this but rather present them as part of the historical record. The system, as mentioned above, clearly indicates that the responses are based on historical archive material.

4

Results

This chapter will present the results of the first six queries about the play Hamlet being asked to the RAG based system (see tables A.1 & B.1 for the other six queries not presented here). All the context required to answer the queries existed within the archive to see if the system could answer them correctly by retrieving and interpreting the context correctly. These queries ranged from roles people were playing, queries about the Hamlet sets and the people contributing to them and queries about reviews. Some queries specified the year while others didn't. Three of the queries were short fact based queries while the other three were open queries. Two tables are presented, where different pairs of chunk sizes were tested, where a chunk is a small segment of data. There is a chunk size for the structured data which was web scraped from the rolebook and one for sentence data which were sentences from the different newspaper articles. The following pairs of sizes were tested:

(**structured chunk size_sentence chunk size**)
1_5 (Table 4.1)
2_3 (Table 4.2)

Structured chunk size is the number of JSON objects in each chunk while sentence chunk size is the number of sentences in each chunk. In total 8 chunks were retrieved for each query, where there was no fixed amount of chunks being retrieved from each source. The table will also provide a small subset of the context being retrieved from each query used to verify

the response. This was also done to not overwhelm the result with massive amounts of context that came with each query and instead provide the most relevant context. Not all relevant context will be shown either, as it was sometimes too much to show for some queries.

4.1 Q1 – Q6 Chunk Size 1_5 Result

Table 4.1: Q1 – Q6 chunk size 1_5 result

Name	**Query**	**Response**	**Relevant context**
Q1	Vem spelade som Hamlet 1960?	Enligt den givna kontexten spelade Ulf Palme rollen som Hamlet i uppsättningen år 1960.	”Hamlet” gavs i Alf Sjöbergs regi och med Ulf Palme i titelrollen och Inga Tidblad som drottningen, Bengt Eklund som kungen, Anders Henrikson som Polonius och Christina Schollin som Ofelia.
Q2	Vem regisserade Hamlet uppsättningen 2008?	Enligt den givna kontexten regisserade Staffan Valdemar Holm uppsättningen Hamlet år 2008.	Typ: Yrke År: 2008 Person: Staffan Valdemar Holm Yrke: Regissör Nyckel: Regissör 2008
Q3	På vilken scen spelades Hamlet uppsättningen 1974?	Uppsättningen Hamlet år 1974 spelades på Stora scenen. Den hade premiär den 11 april 1974 och bestod av 44 föreställningar.	Titel: Hamlet Scen: Stora scenen Antal föreställningar: 44 Premiärdatum: 1974-04-11

Table 4.1 – continued from previous page

Name	Query	Response	Relevant context
Q4	Hur bidrog Alf Sjöberg till Hamlet pjäsen?	Alf Sjöberg bidrog till uppsättningen av Hamlet genom att undvika romantisk sentimentalitet och melodramatik. Han skapade ett drama om människan i hennes höghet och förnedring, istället för ett krönikespel. Sjöbergs regi var puristisk och han vänt ryggen åt textkritiken. Han fokuserade på att framföra Shakespeares skådespel på ett sätt som gjorde att alla de gamla slitna bevingade orden levde upp i ny glans och härlighet.	Men först som sist skall det konstateras att Alf Sjöberg har gjort en alltigenom ståtlig ”Hamlet”, utan alla konstigheter. Sjöbergs regi kan närmast kallas puristisk och han har i det närmaste vänt ryggen åt textkritiken.

Table 4.1 – continued from previous page

Name	Query	Response	Relevant context
Q5	Vad tyckte Martin Strömberg om uppsättningen av Hamlet 1960?	Martin Strömberg tyckte att uppsättningen av Hamlet år 1960 stod fri gentemot föregångarna. Han ansåg att den inte blev ett krönikespel, utan ett drama om människan i hennes höghet och förnedring. Strömberg uppskattade att regin undvek både romantisk sentimentalitet och melodramatik.	Martin Strömberg säger i sin recension i ST bl a: ”Denna Shakespeares ödestragedi står fri gentemot föregångarna, i Alf Sjöbergs händer har den inte blivit ett krönikespel, utan ett drama om människan i hennes höghet och förnedring”.

Table 4.1 – continued from previous page

Name	Query	Response	Relevant context
Q6	Hur beskrevs Ulf Palmes skåde-spelande av Hamlet?	Ulf Palmes skåde-spelande av Hamlet beskrevs som en su-verän prestation. Han framställdes som en mogen och reflek-terad människa, en upplyst furste som kommit till insikt om människans ansvar och skyldigheter. Hans Hamlet var inte gåtfull eller komplicerad, utan väntade helt enkelt på det rätta tillfället att agera. Palmes tolkn-ing av Hamlet skiljde sig från traditionella roller som stormande hero eller bindgalen romantiker. Han gav en ny dimen-sion till karaktären, fri från romantisk sentimentalitet och melodramatik.	Ulf Palmes Hamlet gav en sådan utvidgning av kunskapen. Han var ingen stormande hero, ingen bindgalen romantiker,lyftande dödskallen med höga gester, ingen trånande yngling i uppror, in-gen cyniker, ingen drömmare, över-huvudtaget ingen ”typ”,som dominerar i kraft av sin ensidighet.

4.1.1 Additional Notes

Q1 (Fact based query): The most relevant context was retrieved from the newspaper articles and was interpreted correctly to create a **factual re-sponse**.

Q2 (Fact based query): The most relevant context was retrieved from the Royal Dramatic Theatre's rolebook website and was interpreted correctly to create a **factual response**.

Q3 (Fact based query): The most relevant context was retrieved from the Royal Dramatic Theatre's rolebook website and was interpreted correctly to create a **factual response**.

Q4 (Open query): Not all relevant context is being shown in the table but most of it came from reviews in articles which means that the response was based on reviewers perspective. The context was interpreted correctly which can be considered a **factual response** depending on if it should be through reviewers perspectives or pure facts which the question did not specify.

Q5 (Open query): Context from the correct reviewer was retrieved from the article text and was interpreted correctly to create a **factual response**

Q6 (Open query): Not all relevant context is being shown in the table but most of it came from reviews in articles. The context was used correctly to create a **factual response**

4.2 Q1 – Q6 Chunk Size 2_3 Result

Table 4.2: Q1 – Q6 chunk size 2_3 result

Name	Query	Response	Relevant context
Q1	Vem spelade som hamlet 1960?	Ulf Palme spelade rollen som Hamlet i uppsättningen år 1960.	”Hamlet” gavs i Alf Sjöbergs regi och med Ulf Palme i titelrollen och Inga Tidblad: som drottningen,Bengt Eklund som kungen,Anders Henrikson som Polonius och Christina Schollin som Ofelia.

Table 4.2 – continued from previous page

Name	Query	Response	Relevant context
Q2	Vem regisserade hamlet uppsättningen 2008?	Uppsättningen Hamlet 2008 regisserades av Staffan Valdemar Holm.	Typ: Yrke År: 2008 Person: Staffan Valdemar Holm Yrke: Regissör Nyckel: Regissör 2008
Q3	På vilken scen spelades hamlet uppsättningen i 1974?	Jag kunde inte hitta någon information om uppsättningen Hamlet år 1974 i den givna kontexten. Kan du omformulera frågan?	

Table 4.2 – continued from previous page

Name	Query	Response	Relevant context
Q4	Hur bidrog Alf Sjöberg till hamlet pjäsen?	Alf Sjöberg bidrog till Hamlet-pjäsen genom att skapa en uppsättning som fokuserade på dramat som ett drama om människan, snarare än ett krönikespel. Han behöll alla scener och de sköna bevingade orden, men gjorde inga väsentliga strykningar. Sjöbergs version av Hamlet var en protest mot filologisk vivisektion och psykologisk djupdykning. Han förlade dödgrävarscenen till vintern, vilket gjorde scenen ruffigare. Sjöbergs regi bidrog till att göra uppsättningen av Hamlet till en märkesdag i nationalscenens historia.	Det mest uppseendeväckande infallet i hela föreställningen är att dödgrävarscenen förlagts till vintern, med istappar av plast på ett dekorativt gravkors och projicierad snöyra i fonden. Det är möjligt att den meteorologiska omdiktningen gör scenen ruggigare,Inen arrangemanget förutsätter dessvärre att Ofelia stätt lik i bortåt sex månader. Bortset från en sådan olyckshändelse är det emellertid lätt att uppfatta Sjöbergs ”Hamlet” som en protest mot filologisk vivisektion och psykologisk djupdykning.

Table 4.2 – continued from previous page

Name	Query	Response	Relevant context
Q5	Vad tyckte Martin Strömberg om uppsättningen av hamlet i 1960?	Martin Strömberg tyckte att uppsättningen av Hamlet år 1960 stod fri gentemot föregångarna och att den inte blev ett krönikespel, utan ett drama om människan i hennes höghet och förnedring. Han ansåg att Alf Sjöbergs regi gav stycket en ny dimension.	Martin Strömberg säger i sin recension i ST bl a: ”Denna Shakespeares ödestragedi står fri gentemot föregångarna, i Alf Sjöbergs händer har den inte blivit ett krönikespel, utan ett drama om människan i hennes höghet och förnedring”.

Table 4.2 – continued from previous page

Name	Query	Response	Relevant context
Q6	Hur beskrevs Ulf palmes skåde-spelande av hamlet pjäsen?	Ulf Palmes skåde-spelande i Hamlet beskrevs som en su-verän prestation med våldsam kraft och klarhet. Han tolkade Hamlet som en real-istisk och viril gestalt, långt ifrån den tradi-tionella romantiska eller gåtfulle karak-tären. Palmes Hamlet var ingen stormande hero eller trånande yngling, utan en komplex och psykolo-giskt motiverad figur. Hans tolkning gav en ny dimension till rollen och utvidgade kunskapen om karak-tären. Kritiker som Ebbe Linde och Mar-tin Strömberg hyllade hans insats.	Om Ulf Palmes insats skriver bl a Ebbe Linde: 'Det är en suverän prestation. Inte fråga om annat. Hamlet gestaltas i Alf Sjöbergs regi av Ulf Palme i en rollanalys som tagit fasta på de virila sidorna hos danaprinsen. Palmes Hamlet står närmare realistisk teater än en romantisk tolkning.

4.2.1 Additional Notes

Q1 (Fact based query): The most relevant context was retrieved from the newspaper articles and was interpreted correctly to create a **factual response**.

Q2 (Fact based query): The most relevant context was retrieved from the Royal Dramatic Theatre's rolebook website and was interpreted correctly to create a **factual response**.

Q3 (Fact based query): No relevant context was retrieved from the archive. The archive contains this information, which means that there was a **retrieval failure**. The response tells the user that it was not able to find any information on this topic (**successful abstention**) and asks the user to formulate a new question.

Q4 (Open query): The most relevant context was retrieved from the newspaper articles and was interpreted correctly to create a **factual response**.

Q5 (Open query): The most relevant context was retrieved from the newspaper articles and was interpreted correctly to create a **factual response**.

Q6 (Open query): The most relevant context was retrieved from the newspaper articles and was interpreted correctly to create a **factual response**.

4.3 Context Retrieval Hit Rate Summary

While Tables 4.1 and 4.2 present selected examples, the context retrieval performance for all queries is summarized in Table 4.3 below. To view the rest of the queries, see Tables A.1 & B.1. Query hit rate refers to whether the system retrieved context relevant to the query, even if the generated response contained minor contextual misinterpretations. The chunking configuration (structured chunksize_sentence chunksize) 1_5 performs better than 2_3 for fact-based queries, but for open-ended queries they got the same score.

Table 4.3: Summary of retrieval hit rate for fact-based and open-ended queries.

Chunk configuration	**Fact-based query hit rate**	**Open-ended query hit rate**	**Total**
1_5	5/6 (83%)	5/6 (83%)	10/12 (83%)
2_3	3/6 (50%)	5/6 (83%)	8/12 (67%)

4.4 Final Chatbot Prototype

The chat interface enables the user to type in a question and receive a response, see Figure 4.1. The user's natural language query and the chunks are translated into numerical vectors. The query vector is then compared to all the chunk vectors to retrieve the top 8 most semantically similar matches. These 8 chunks are used as context and then processed by the AI model to generate a grounded response. To avoid false information in the response, the system is instructed to prioritize honesty, which means informing the user of missing data rather than fabricating information. However, there is a risk that the AI model misunderstands the given context and therefore responds with incorrect information. The system is also instructed to ask the user to formulate a new question when it did not find relevant context, because a reformulated question might result in a more successful retrieval.



Figure 4.1: Screenshot of the Hamlet AI chatbot.

5

Discussion

The results of this study are discussed in relation to the research question of how a locally run AI model can improve access to the Royal Dramatic Theatre's archive.

5.1 Data Chunking

This study investigates how different chunk sizes affect retrieval and response quality. The results show that the system can often retrieve relevant context and generate accurate responses for both open-ended and fact-based queries. However, the result varies depending on the type of data and the chosen chunk size. This shows the importance of adapting the chunking strategy to the structure of the data.

5.1.1 The Impact of Chunk Size on Different Data Types

The results indicate that the chunk size has an impact on retrieval performance, particularly for fact-based queries. As shown in Table 4.3, the configuration with smaller structured chunks (1_5) outperforms the larger structured chunks (2_3) when answering factual queries.

The relevant context for fact-based queries is often data from the rolebook. The rolebook data is represented in a JSON format where one item represents one fact, such as what actor played a certain role. This means that a chunk size of 1 contains one fact and a chunk size of 2 contains two facts. When two or more facts are in the same chunk, the semantic meaning

is diluted, which makes retrieval less precise. This is likely the reason as to why smaller chunk sizes are more effective for structured fact-dense data. These findings are consistent with previous research, such as the FolkRAG study, which shows that a smaller chunk size is better suited for archival material with discrete descriptive elements, to not lose semantic specificity (Kelly et al., 2025).

For open-ended queries, the relevant context is often data from unstructured article text. Each chunk contains a number of sentences, in this case 5 and 3. Compared to the rolebook data, the article sentences are not as fact-dense and uses more descriptive words. Both chunking configurations performed similarly, as seen in Table 4.3. With larger chunks, the system receives more cohesive context, while smaller chunk sizes might cut off important context from the article. In this study, the difference between the two chunk sizes did not show a clear winner. This suggests that open-ended queries are less sensitive to chunk size compared to fact-based queries. However, it might be more difficult to generate a general and cohesive response with a very small chunk size.

5.2 Other Findings

Other findings cover anything outside the effects of different chunk sizes. The results also showcased similarities in response and how context was retrieved and interpreted regardless of chunk size.

5.2.1 Contextual Misinterpretation

Regardless of chunk size, there were cases where the retrieved context got misinterpreted. This occurred with more specified open queries where variations of the same query had their context interpreted differently. For example Q11 and Q12 in table A.1 & B.1 are similar since they ask about reviews about the hamlet performances from 1960. The difference is that Q11 doesn't ask about any specific reviews while Q12 asks about a specific newspaper's reviews. While similar context was being retrieved for both queries, the model was misinterpreting it in it's responses to Q12.

The model assumed that all context being retrieved was relevant to the query and in this case it assumed that all reviews came from the newspaper DN. The model was instructed to not give an answer without the proper context, however this rule gets broken when it thinks that the correct type

of context will always be relevant to the specific query. This type of contextual misinterpretation aligns with findings from the FolkRAG study, which notes that limitations in retrieval precision can lead to hallucination issues when the model struggles to handle irrelevant or ”noisy” information in the retrieved context (Kelly et al., 2025).

5.2.2 Missing the Correct Context

Chunk size does not have an effect on what similarity score each chunk will receive and is instead dependent on the query and the available data. The embedding process struggled sometimes with finding relevant chunks for fact based queries that mentioned the word ”Hamlet“ such as Q7 in table A.1 & B.1. A reason for this is that a lot of the structured rolebook data contained the word ”Hamlet“. Without proper metadata for the structured data, the outcome of the embedding process suffers as there was no proper way to differentiate between the role and the play ”Hamlet“. JSON objects from the rolebook regarding the play Hamlet (scene played on, number of performances, release date, etc..) were being retrieved instead when queried on who played the role as Hamlet in 1986?. The word ”Hamlet“ was being matched without any other added context which gave the wrong JSON objects a higher similarity score because they also mentioned the word ”Hamlet“.

Earlier studies of semantic search discuss how such ”dual interpretations“ are a problem when relying on conventional keyword retrieval methods (Reddy K et al., 2025). Reddy K et al. (2025) proposed context-aware embeddings, where the transformer models Sentence-BERT and MPNet were used together to create contextual embeddings. The study showed potential in understanding hidden meaning in users queries and understanding language patterns to avoid word ambiguity (Reddy K et al., 2025). Context aware embeddings were not used within the Hamlet RAG system and it can be considered potential future research.

5.3 Comparison to Previous Research

Earlier research such as the FolkRag study suggest that a smaller chunk size is better for material with discrete descriptive elements (Kelly et al., 2025). Discrete in this context means elements that present distinct units of information. The structured data in this study can be considered to be this

type of information. Things like date, role and job were some of these elements. The smaller structured size (1_5) showed a higher query hit rate for fact-based queries where the structured context was used. This result corresponds to the FolkRag study's result where a smaller chunk size for such data was proven to be more effective for context retrieval.

This study also aligns with previous research showing that retrieval limitations and noisy context can contribute to hallucination issues and contextual misinterpretation in the generated responses (Kelly et al., 2025).

The retrieval issues caused by the double meaning of the word "Hamlet" can also be related to previous research. Studies have shown that dual interpretations can negatively affect retrieval quality when the system lacks contextual understanding (Reddy K et al., 2025). This suggests that more context-aware embedding techniques could improve retrieval performance in archival RAG systems.

5.4 Future Research

5.4.1 Metadata

The only metadata the system used was for the article text. More specifically each article chunk starts with "(Text from different articles about the play Hamlet year 1960)". Future research could study how added metadata affects the retrieval and interpretation of context. For example, tagging the article chunks with metadata regarding what newspaper published the article and at what date. This could potentially solve the issue with how the model assumed all correct type of context being retrieved was also relevant to the specific query. It could also create more transparency as having metadata for chunk sources could make it easier for the model to present what sources the response was based on. The structured data could also be tagged with metadata to give context to each JSON object instead of relying on the embedding process's word matching, which caused issues with the word "Hamlet" in some cases.

5.4.2 Query Transformation

Something that would considerably make the system better would be to improve context retrieval, because the system tends to generate good responses when given relevant context. The context is retrieved by comparing the semantic meaning between the query and the chunks. This means that the query is crucial for context retrieval. Two similar but differently formulated queries can therefore retrieve different chunks for context. It is however not ideal to rely on the user to formulate queries in a way that will retrieve relevant context. A technique that could be implemented to improve this issue is Query Transformation, where the LLM generates a number of new, reformulated queries to increase the probability of retrieving relevant chunks. Another way of doing this is to have the LLM first generate a hypothetical answer to the query, and then send the answer to retrieve context instead of the query.

Simple preprocessing steps such as spellchecking of the query could also improve retrieval performance. Since the system relies on semantic similarity between query and chunks, a misspelled word in the query may significantly alter the query representation and lead to failed retrieval. Correcting the spelling errors by having the LLM reformulate the query could therefore increase the likelihood of retrieving relevant context.

5.4.3 Improve System Instructions

The system is given a set of predefined instructions that control how the LLM generates responses. These instructions primarily aim to ensure that the responses are grounded in the retrieved context and that there is no fabricated information. Also, the system is instructed to inform the user when no relevant context was found and to reformulate the question.

In Table 4.2, the response to Q3 shows that the system is able to follow the given instructions about informing the user when no context was found. Therefore, these instructions are crucial for minimizing the risk of information fabrication (hallucinations). However, this could lead to the system not utilizing partially relevant context, to avoid not following the instructions perfectly. Future improvements could involve refining the system instructions to find a balance between giving clear instructions but not being too strict. Also, more guidance could be added to get more tailored responses.

Post-Evaluation Instruction Refinements

After the evaluation was completed, alternative system instructions were tested to explore whether the model's behavior could be improved. These revised instructions were structured as a bullet-point list, with additional emphasis on the fact that all retrieved context is not guaranteed to be relevant to the query. This improved the issue discussed in Section 5.2.1, where the model incorrectly assumed that retrieved review text originated from the newspaper DN, despite the context not explicitly stating this. Preliminary testing indicated that the revised instructions reduced such contextual misinterpretations. Instead of generating an inaccurate answer, the model informed the user that it could not find information on this. This suggests that more explicit system instructions may improve the model's ability to critically evaluate retrieved context.

5.5 Use of IT Tools

Several IT tools were used during the implementation and evaluation of the system. Python was chosen as the primary programming language because of its widespread use in AI development and the availability of relevant libraries. Ollama enabled the local deployment of the LLM, and Mistral Small was selected because it is a European model and it performed well in response generation. PaddleOCR was used for transcription of images of the historical newspapers, particularly because of its ability to handle column-based layouts. Streamlit was used to develop the prototype interface, as its Python-based framework simplified integration with the rest of the system. ChatGPT was used for improving system instructions and troubleshooting certain implementation issues.

5.6 Societal Implications

This study highlights the potential of using RAG-based AI systems to make historical archives more accessible through natural language interaction. Such systems could lower the barrier to exploring archival material without prior knowledge of archival systems, and make large collections more searchable for archivists. In the long term, this may contribute to increased public access to historical archives.

At the same time, the study also demonstrates limitations of archival AI systems, such as AI hallucinations that can misrepresent the archival material. If users place too much trust in generated responses without verifying the original source, there is a risk of spreading incorrect interpretations of historical documents. Therefore, transparency remains important when implementing AI systems.

5.7 Conclusion

The aim of this study was to investigate how a locally run AI model can make the Royal Dramatic Theatre's archive more accessible. The results indicate that accessibility can be improved by allowing users to query archival data using natural language and receive grounded responses based on retrieved archival context. The system was generally able to retrieve relevant context and generate accurate answers for both fact-based and open-ended queries, demonstrating how a locally deployed AI model can function as an interface for exploring archival material without the need for prior knowledge of the archive system.

However, the findings also highlight some limitations. The performance of the system depends heavily on the quality of the context retrieval process, which is influenced by factors such as chunk size, data structure and query formulation. While smaller chunk sizes improved precision for structured data, open-ended queries were less sensitive to chunk size. These results suggest that an optimal chunking strategy should not be uniform across all datatypes, but instead tailored to the structure and intended use of the data. In addition, issues such as contextual misinterpretation indicate that the system is not yet fully reliable and needs refinements.

Overall, the study shows that a locally run AI model can make archival data more accessible through natural language interaction, retrieval-based grounding and adequate preprocessing of archival data. The findings emphasize that the effectiveness of such systems depends on the system design, including data structuring, chunking strategies and instruction tuning. With further improvements, systems of this kind have the potential to make cultural heritage archives more accessible to a broader audience.

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A

Q7 – Q12 Chunk Size 1_5 Result

Table A.1: Q7 – Q12 chunk size 1_5 result

Name	Query	Response	Relevant context
Q7	Vem hade rollen som hamlet 1986?	Jag kunde inte hitta någon information om vem som hade rollen som Hamlet i uppsättningen år 1986. Kan du omformulera frågan?	
Q8	Vem var scenograf för hamlet 1942?	Scenografen för uppsättningen Hamlet år 1942 var Sven-Erik Skawonius.	Typ: Yrke År: 1942 Person: Sven-Erik Skawonius Yrke: Scenograf Nyckel: Scenograf 1942

Table A.1 – continued from previous page

Name	Query	Response	Relevant context
Q9	Vem spelade som Polonius 1974?	Sigge Fürst spelade rollen som Polonius i uppsättningen Hamlet år 1974. Denna information finns tydligt angiven i den givna kontexten.	Typ: Roll Skådespelare: Sigge Fürst Roll: Polonius Beskrivning: Datum: 1974-04-11 År: 1974 Nyckel: Polonius 1974
Q10	Hur bidrog per myberg till hamlet pjäsen?	Per Myrberg bidrog till uppsättningen Hamlet år 1960 genom att spela rollen som Laertes. Hans insats var en del av den övergripande enheten som gjorde att föreställningen upplevdes som en helhet. Myrbergs prestation smälte samman med de övriga skådespelarnas insatser och bidrog till att skapa en överväldigande upplevelse för publiken.	Alf Sjöbergs Hamlet på Dramatens stora scen — forsta forestalmigen P skådespelarna i rollistan. Ulf Palme gjorde Hamlet, Inga Tidblad Hamlets mor, Bengt Eklund kung Klaudius och Anders Henrikson Polonius. Ofelia gjordes av en dramatenelev Christina Sehollin, Laertes av Per Myrberg och vidare förekom många goda skådeoeaCvsaes Om Ulf Palmes insats skriver bl a Ebbe Linde: 'Det är en suverän prestation.
Q11	Hur var recensionerna för hamlet uppsättningen i 1960?	Jag kunde inte hitta någon information om recensionerna för uppsättningen Hamlet år 1960 i den givna kontexten. Kan du omformulera frågan?	

Table A.1 – continued from previous page

Name	Query	Response	Relevant context
Q12	Vad skrev tidningen DN om uppsättningen av hamlet i 1960?	Tidningen DN skrev om uppsättningen av ”Hamlet” år 1960 att Ulf Palmes tolkning av Hamlet inte var en typisk hero eller romantiker, utan en mogen och reflekterad människa. Regissören Alf Sjöbergs styrka låg i den energiska rytmen och den laddade dramatiken, även om den energiska rytmen gick förlorad i styckets första scener. Uppsättningen förankrades till grå vikingatid med en nordisk accentuering genom vinterstämning och frostigt norrsken. Den klassiska översättningen av A. Hagberg användes, och dödgrävarscenen var förlagd till vintern.	Ulf Palmes Hamlet gav en sådan utvidgning av kunskapen. Han var ingen stormande hero, ingen bindgalen romantiker,lyftande dödskallen med höga gester, ingen trånande yngling i uppror, ingen cyniker, ingen drömmare, överhuvudtaget ingen ”typ”,som dominerar i kraft av sin ensidighet De har åstadkommit en fyndig lösning genom att förankra dramat till grå vikingatid och gett det en nordisk accentuering genom vinterstämning och frostigt norrsken,med dödgrävarscenen förlagd till vintern med Facklan var ett viktigare vapen än floretten i Ulf Palmes hand.

Q7 (Fact based query): No relevant context was retrieved from the archive. The archive contains this information, which means that there was a **retrieval failure**. The response tells the user that it was not able to find any information on this topic (**successful abstention**) and asks the user to formulate a new question.

Q8 (Fact based query): The most relevant context was retrieved from the Royal Dramatic Theatre's rolebook website and was interpreted correctly to create a **factual response**.

Q9 (Fact based query): The most relevant context was retrieved from the Royal Dramatic Theatre's rolebook website and was interpreted correctly to create a **factual response**.

Q10 (Open query): The most relevant context contains some correct information where the actors role being played is described, however the model is also making assumptions on what effect Per Myrbergs performance had on the crowd which wasn't explicitly told in the context. This creates a **Partially contextual misrepresentation response**.

Q11 (Open query): No relevant context was retrieved from the archive. The archive contains this information, which means that there was a **retrieval failure**. The response tells the user that it was not able to find any information on this topic (**successful abstention**) and asks the user to formulate a new question.

Q12 (Open query): There is context from reviews being retrieved which is relevant, however there is no mention in the context that the reviews are from the newspaper DN and the model is making assumptions that it is. This ends up creating a **Contextual misrepresentation response**.

B

Q7 – Q12 Chunk Size 2_3 Result

Table B.1: Q7 – Q12 chunk size 2_3 result

Name	Query	Response	Relevant context
Q7	Vem hade rollen som hamlet 1986?	Jag kunde inte hitta någon information om vem som hade rollen som Hamlet 1986. Kan du omformulera frågan?	
Q8	Vem var scenograf för hamlet 1942?	Scenografen för uppsättningen Hamlet år 1942 var Sven-Erik Skawonius.	Typ: Yrke År: 1942 Person: Sven-Erik Skawonius Yrke: Scenograf Nyckel: Scenograf 1942

Table B.1 – continued from previous page

Name	Query	Response	Relevant context
Q9	Vem spelade som Polonius 1974?	Jag kunde inte hitta någon information om vem som spelade Polonius 1974 i den givna kontexten. Kan du omformulera frågan?	
Q10	Hur bidrog per myberg till hamlet pjäsen?	Jag kunde inte hitta någon information om Per Myberg i den givna kontexten. Kan du omformulera frågan?	

Table B.1 – continued from previous page

Name	Query	Response	Relevant context
Q11	Hur var recensionerna för hamlet uppsättningen i 1960?	Recensionerna för Hamletuppsättningen 1960 var blandade. Alf Sjöbergs regi framhölls för sin konsekvens och kraft, men kritiker menade att romantiken och psykologiska djupen undertrycktes. Inga Tidblads tolkning av Gertrud beskrevs som värdig och skuldlös, medan Bengt Eklunds Claudius och Anders Henriksons Polonius uppmärksammades för sina accentuerade roller. Christina Schollin hyllades för sin Ofelia, som skilde sig från traditionen. Samtidigt ansågs uppsättningen befriad från tragisk spänning.	Alf Sjöbergs regi vad som är krönikespel i ”Hamlet”, allt det som förenar stycket med författarens Henrikar och Rickardar.Vad som har blivit försummat är romantiken, ljusdunklet, poesin, de berömda psykologiska djupen och gåtorna; ja, man kan säga att allt detta har blivit planmässigt undertryckt.

Table B.1 – continued from previous page

Name	Query	Response	Relevant context
Q12	Vad skrev tidningen DN om uppsättningen av hamlet i 1960?	Tidningen DN skrev att Alf Sjöberg och dekoratören Lennart Mörk gav en brett upplagd och fullständig version av Shakespeares drama ”Hamlet” i August Hagbergs klassiska översättning. De skapade en miljö som förankrades i vikingatiden med inslag av renässans i dräkter och rekvisita. Ulf Palmes tolkning av Hamlet var inte en typisk hero eller romantiker, utan en mer nyanserad och komplex karaktär. Inga Tidblad spelade drottningen med ett allvarsdrag och ett intryck av skuldlöshet. Regin undertryckte romantiken och poesin för att framhålla krönikespelens aspekter. Bengt Eklund och Anders Henriksons roller som kungen respektive Polonius gav en accent av list och infamitet.	Ulf Palmes Hamlet gav en sådan utvidgning av kunskapen. Han var ingen stormande hero, ingen bindgalen romantiker,lyftande dödskallen med höga gester, ingen trånande yngling i uppror, ingen cyniker, ingen drömmare, överhuvudtaget ingen ”typ”,som dominerar i kraft av sin ensidighet De har åstadkommit en fyndig lösning genom att förankra dramat till grå vikingatid och gett det en nordisk accentuering genom vinterstämning och frostigt norrsken,med dödgrävarscenen förlagd till vintern med Facklan var ett viktigare vapen än floretten i Ulf Palmes hand.

Q7 (Fact based query): No relevant context was fetched from the archive. The archive contains this information, which means that there was a **retrieval failure**. The response tells the user that it was not able to find any information on this topic (**successful abstention**) and asks the user to formulate a new question.

Q8 (Fact based query): The most relevant context was fetched from the Royal Dramatic Theatre's rolebook website and was interpreted correctly to create a **factual response**.

Q9 (Fact based query): No relevant context was fetched from the archive. The archive contains this information, which means that there was a **retrieval failure**. The response tells the user that it was not able to find any information on this topic (**successful abstention**) and asks the user to formulate a new question.

Q10 (Open query): No relevant context was fetched from the archive. The archive contains this information, which means that there was a **retrieval failure**. The response tells the user that it was not able to find any information on this topic (**successful abstention**) and asks the user to formulate a new question.

Q11 (Open query): Context was fetched from different review sources from the newspaper articles to create a **factual response**.

Q12 (Open query): There is context from reviews being fetched which is relevant, however there is no mention in the context that the reviews are from the newspaper DN and the model is making assumptions that it is. This ends up creating a **Contextual misrepresentation response**.